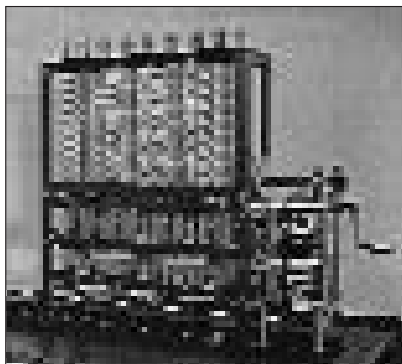


Nothing beyond this prototype segment was actually built.

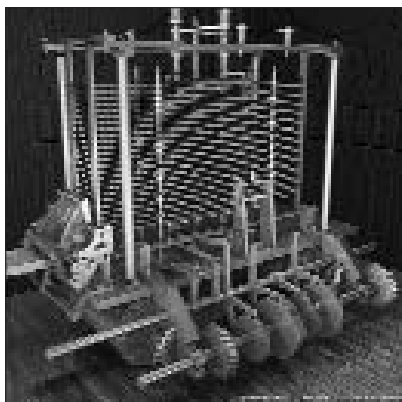
1834

This was the year of the conception and design of the Analytical Engine. The program was stored on punch cards. The inventor continued work on his design for many years; the machine would operate on 40-digit numbers, which is, come to think of it, huge. He called the CPU the “mill,” and the memory he called, pretty obviously, the “store,” which would hold a hundred numbers. Babbage’s plans for his Analytical Engine were complex: several punch card readers, both for programs and data. There was also provision for *conditional jumps*—as in “if-then” statements—on this *mechanical* device! As if that weren’t enough, the *meaning* of the instructions would be interpreted depending on the positioning of metal studs in a slotted barrel (all these words to give you an idea of the complexity). At the end of it, the machine would do an addition in three seconds and a multiplication (or division) in two to four minutes.

The Difference Engine project was cancelled in



The all-famous Difference Engine: a modern interpreted drawing



This image, of a model of the Analytical Engine, won an honourable mention for technical merit in the March/April '99 round of the Internet Raytracing competition